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1. INTRODUCTION

In the context of increasing global concern about energy sustainability and resource management, accurate forecasting of renewable energy production has become a key tool for strategic planning. In this project, we have been contracted by the company U.S. Energy Information Administration (EIA) to develop a predictive model for total renewable energy production in the United States.

The main objective of this study is to provide reliable forecasts that can support decision-making processes related to infrastructure planning, resource allocation, and supply management. To this end, we analyze a time series dataset provided by the agency, which records total renewable energy production.

We will try to identify, estimate, and validate appropriate ARIMA models to generate forecasts and understand the behaviour of the series.

2. BASIC DESCRIPTION OF THE DATASET

The dataset analyzed in this project contains the monthly total primary energy consumption in the United States, measured in billions of BTU (British Thermal Units). The data comes from the U.S. Energy Information Administration (EIA), a public agency responsible for collecting and sharing energy-related information.

The series covers over 30 years, from 1990 to 2019. We can see the plot of the time series in the figure below:

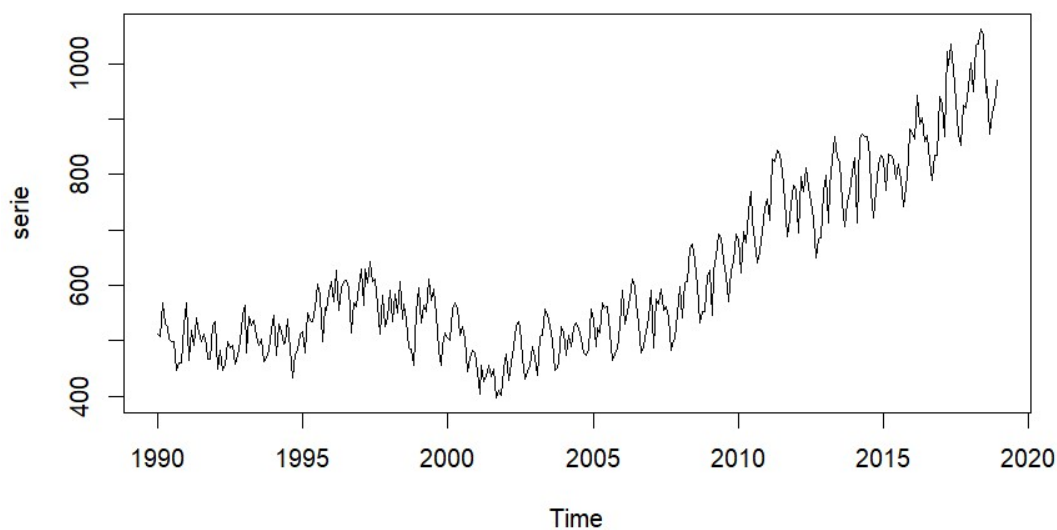


Figure 1. Original time series

In general terms, the data reflects the aggregated energy demand across different sectors, such as residential, industrial, and transportation. With simple look at the plot, a linear positive trend is observed, starting at 2002 until 2019. Also, it seems to be a presence of seasonal patterns, as well as possible changes linked to economic cycles or shifts in energy policies. Those are things that will be taken into account when transformin the series into stationary.

As for data quality, no missing values were found, and the dataset was already provided in a preprocessed form, which makes it easier to work with and ensures good consistency for time series analysis.

3. METHODOLOGY SUMMARY

The methodology applied in this study follows the Box-Jenkins framework for ARIMA modelling, structured into several key stages including identification, estimation, validation, and forecasting.

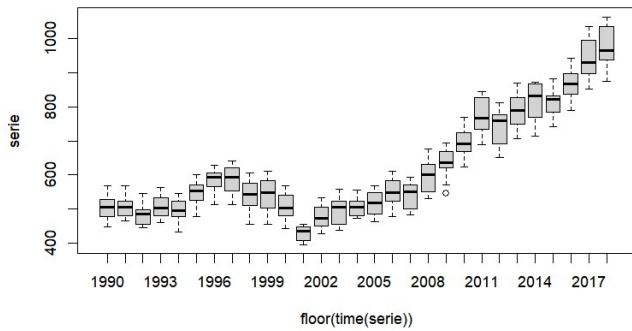
1. The stationarity of the series was assessed using both graphical analysis and statistical tests. When necessary, transformations such as differencing and logarithmic scaling were applied to stabilize the mean and variance of the process.
2. The autocorrelation function (ACF) and partial autocorrelation function (PACF) were analyzed in order to propose candidate ARIMA models. Based on these diagnostics, two model structures were selected.
3. The parameters of the selected models were estimated using R functions, based on standard information criteria such as AIC, BIC, MSE, MAE, MSPE and MAPE.
4. A comprehensive residual analysis was performed to verify the adequacy of the models. This included checking for absence of autocorrelation, normality assumptions and independence.
5. The forecasting performance was evaluated by reserving the last 12 observations of the series as a test set. The best-performing model was then used to generate forecasts for the next 12 months, including prediction intervals.
6. For the final selected model, an automatic outlier detection procedure was applied to interpret the detected anomalies. Once the series was adjusted for outliers, forecasts for the original series were generated using the transformed model and compared with the previous forecasts.

4. RESULTS AND INTERPRETATION

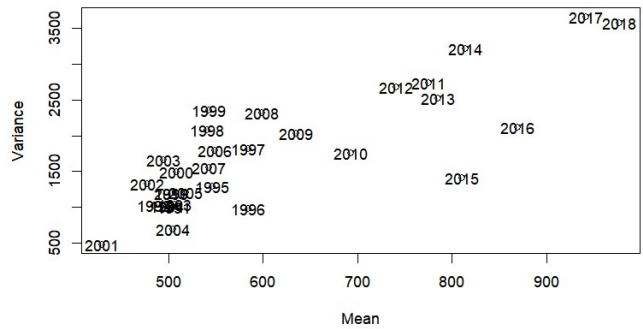
4.1. Identification

To apply ARIMA models, the first step is to ensure that the series is stationary. From Figure 1, we observe that the variability increases with the level of the series, meaning that higher values show more dispersion. This suggests that a logarithmic transformation may help stabilize the variance.

We can confirm this hypothesis using the box plot and the mean and variance plot:

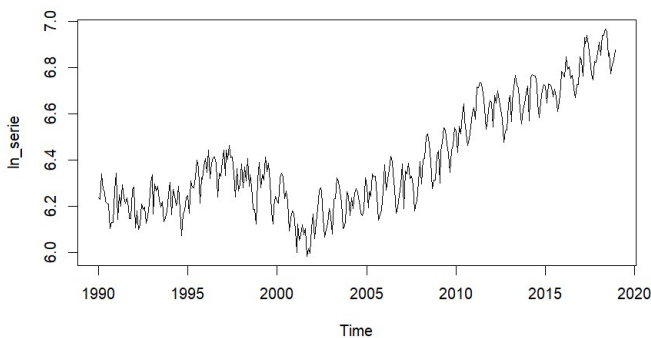


(a) Boxplot of the original time series

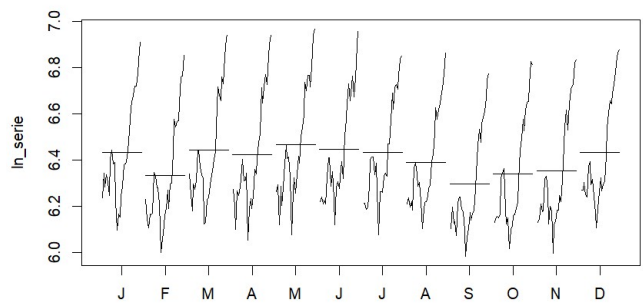


(b) Mean and variance plot

In addition, the original series shows a clear increasing trend and a strong seasonal pattern:

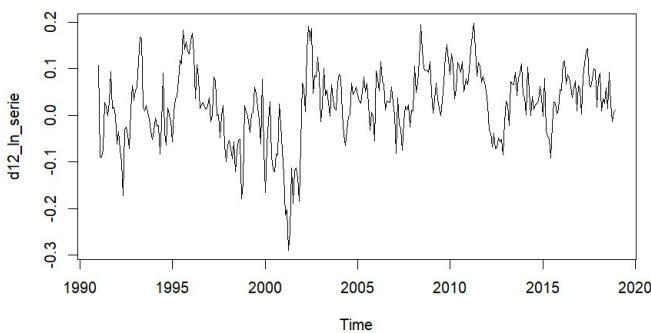


(a) Log-transformed time series

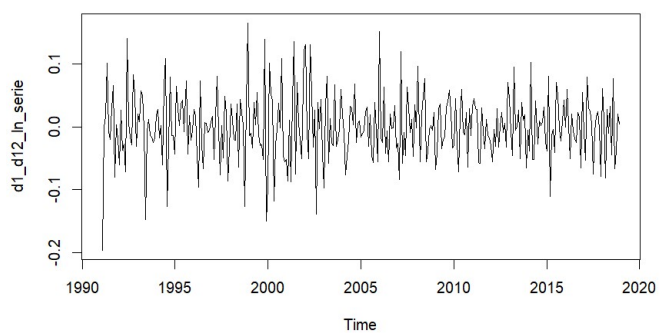


(b) Monthplot of the log-transformed time series

To remove these components, we apply a seasonal difference (lag 12) followed by a regular difference. After these transformations, the series shows a much more stable behaviour.



(a) Log-transformed + Seasonal difference



(b) Log + Seasonal + Regular difference

After applying seasonal and regular differencing, the series becomes approximately stationary in both mean and variance. The final working series, denoted as `energy_wt`, corresponds to series transformation with the lowest variance.

Once stationarity was achieved, the ACF and PACF of the transformed series were analyzed in order to identify potential ARIMA structures.

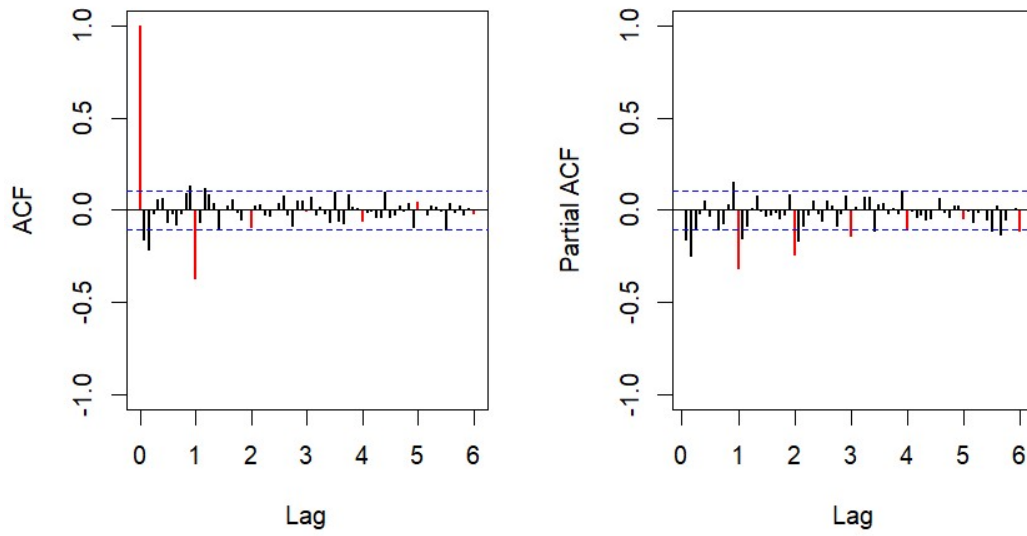


Figure 5. ACF and PACF of energy_wt

The resulting plots suggest the presence of both seasonal and non-seasonal dynamics, with significant seasonal lags (colored in red) and regular lags (colored in black).

4.2. Model identification and selection

Based on the observed autocorrelation structure, four candidate ARIMA models were proposed, combining different configurations of AR and MA components in both the seasonal and non-seasonal parts. These models are:

- **Regular part:**

$$\left. \begin{matrix} \text{AR}(2) \\ \text{MA}(2) \end{matrix} \right\} \Rightarrow \text{ARMA}(1,1)$$

- **Seasonal part:**

$$\left. \begin{matrix} \text{AR}(3) \\ \text{MA}(1) \end{matrix} \right\} \Rightarrow \text{ARMA}(1,1)$$

4 models were proposed choosing the simpler ones in each part:

- **Proposed models:**

$$\text{ARMA}(1,1) + \text{ARMA}(1,1) \Rightarrow \text{ARIMA}(1,1,1)(1,1,1)_{12}$$

$$\text{MA}(2) + \text{MA}(1) \Rightarrow \text{ARIMA}(0,1,2)(0,1,1)_{12}$$

$$\text{ARMA}(1,1) + \text{MA}(1) \Rightarrow \text{ARIMA}(1,1,1)(0,1,1)_{12}$$

$$\text{AR}(2) + \text{MA}(1) \Rightarrow \text{ARIMA}(2,1,0)(0,1,1)_{12}$$

The AIC values and the model summaries are available in the annex.

4.3. Estimation

Based on the AIC values, the following two models were selected for further validation:

• **Model 1:**

– Formula of the model:

$$\log(X_t) \sim \text{ARIMA}(1, 1, 1)(1, 1, 1)_{12}, \quad Z_t \sim N(0, 1.6 \cdot 10^{-3})$$

$$(1 - \phi_1 B)(1 - \Phi_1 B^{12})(1 - B)(1 - B^{12})\log(X_t) = (1 + \theta_1 B)(1 + \Theta_1 B^{12})Z_t$$

$$(1 - 0.49B)(1 - 0.18B^{12})(1 - B)(1 - B^{12})\log(X_t) = (1 - 0.76B)(1 - 0.91B^{12})Z_t$$

- AIC = -1179.53

• **Model 2:**

- Formula of the model:

$$\log(X_t) \sim ARIMA(0, 1, 2)(0, 1, 1)_1 2, \quad Z_t \sim N(0, 1.6 \cdot 10^{-3})$$

$$(1 - B)(1 - B^{12})\log(X_t) = (1 + \theta_1 B + \theta_2 B^2)(1 + \Theta_1 B^{12})Z_t$$

$$(1 - B)(1 - B^{12})\log(X_t) = (1 + 0.05B + 0.06B^2)(1 + 0.05B^{12})Z_t$$

- AIC = -1178.86

The statistical significance of the estimated coefficients was evaluated using t-ratios. All coefficients in the proposed models were found to be significant ($|t| > 2$), so no further simplification was required.

4.4. Validation

4.4.1. Residual analysis

The residual analysis of both models was performed to verify the validity of the model assumptions, including homoscedasticity, normality and independence.

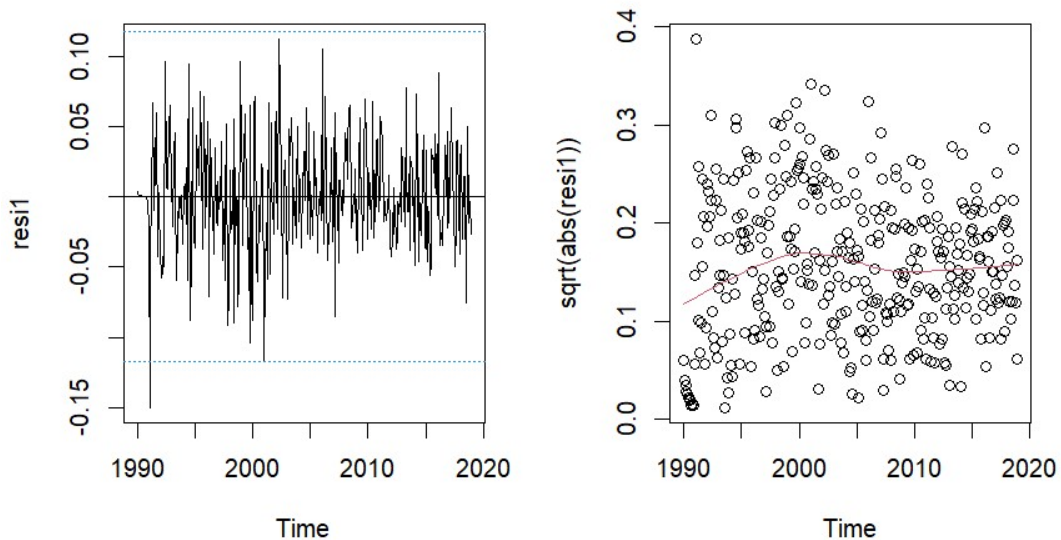


Figure 6. Plot of the residuals of model 1

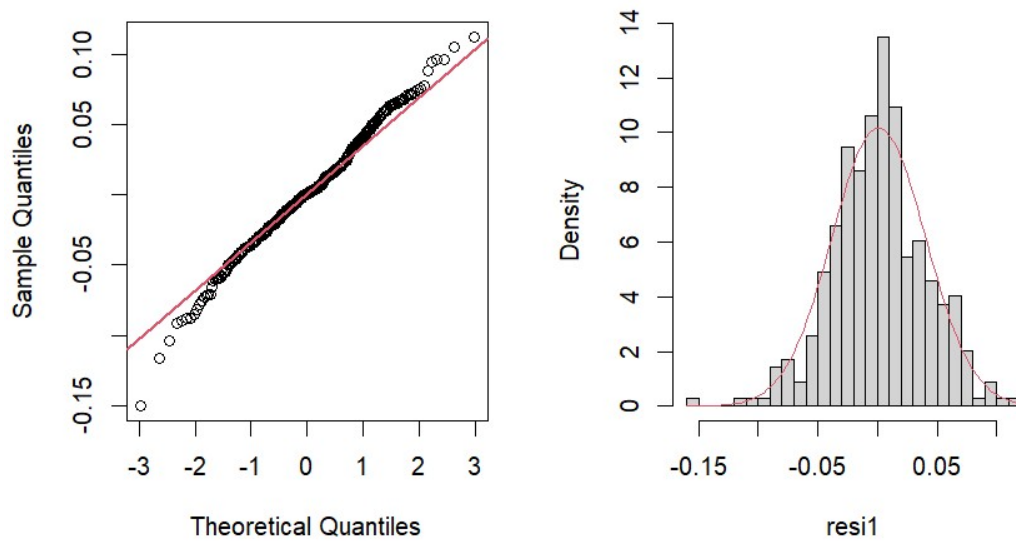


Figure 7. Q-QPlot and Histogram of the residuals of model 1

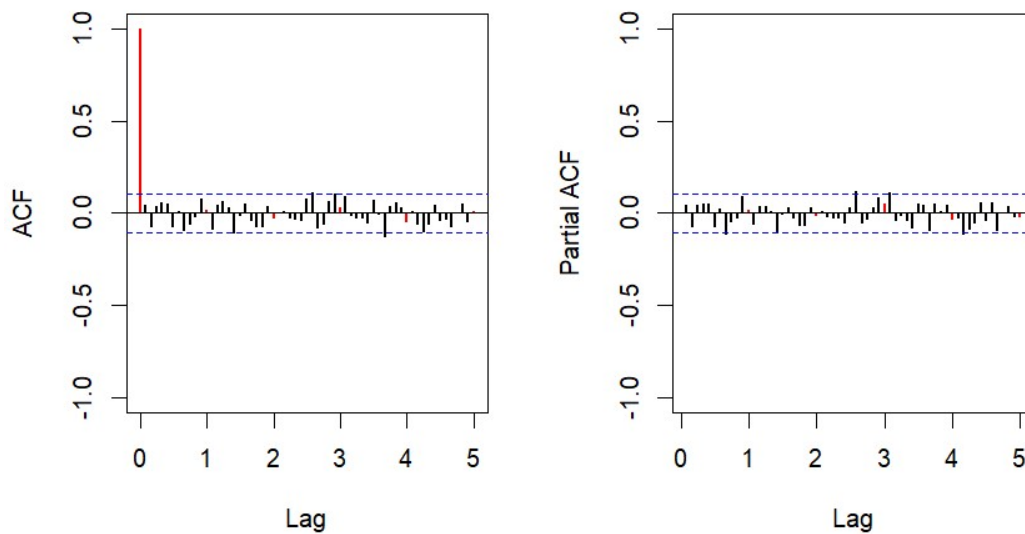


Figure 8. ACF and PACF of the residuals of model 1

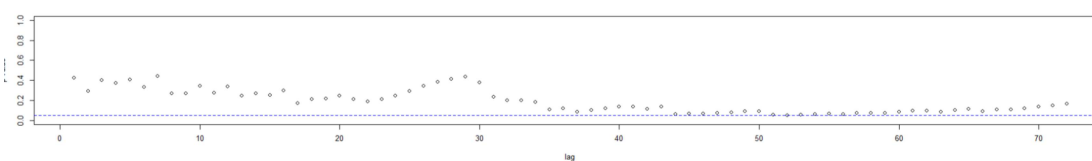


Figure 9. Ljung-Box test plot of model 1

For the first model, evidence of non-constant variance (heteroscedasticity) was detected, particularly in the early part of the series. Additionally, there seems to be an outlier at the beginning.

Regarding normality, the Q-Q Plot of the residuals and the histogram confirm this hypothesis. The first one has most of the points aligned, except for the extremes which might indicate presence of outliers. The histogram is symmetrical and seems to follow a normal distribution.

Additionally, the normality assumption is not rejected according the Shapiro-Wilk test, although small deviations are observed in the tails of the distribution.

The residuals show no significant autocorrelation as confirmed by both the ACF/PACF of residuals. Checking the independency with the Ljung-Box test, some of p-values (points observed in figure 9) are close to the significant limit, but all of them remain above the blue line, so the assumption of independence is validated.

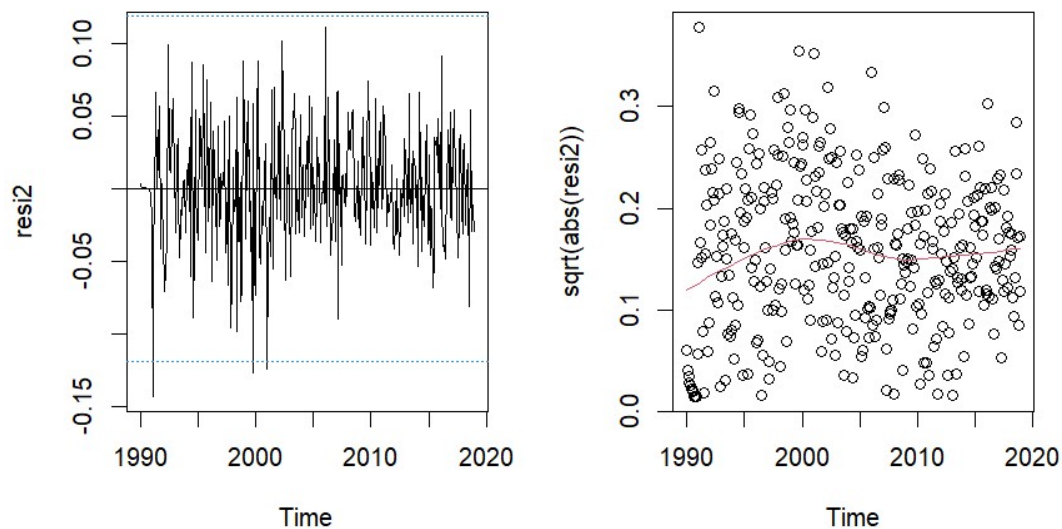


Figure 10. Plot of the residuals of model 2

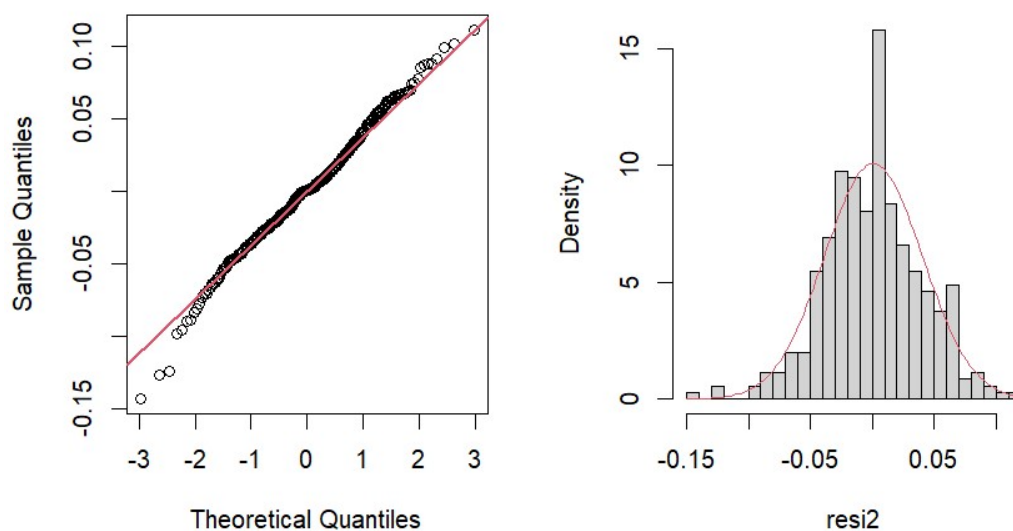


Figure 11. Q-QPlot and Histogram of the residuals of model 2

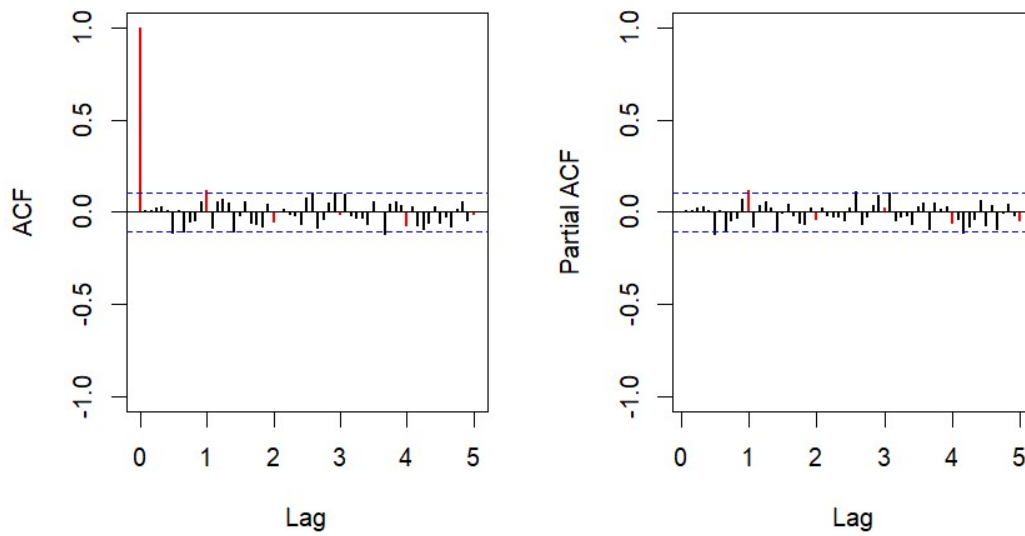


Figure 12. ACF and PACF of the residuals of model 2

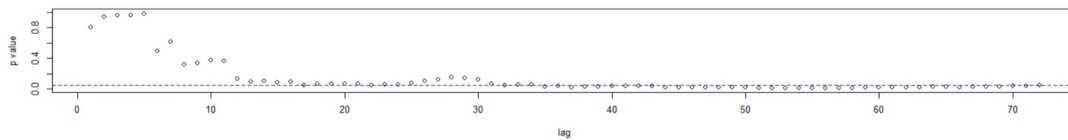


Figure 13. Ljung-Box test plot of model 2

For the second model, residual diagnostics lead to similar conclusions.

In the case of the variance, heteroscedasticity is also observed, but in this case three outliers are detected.

The normality is still validated; the Q-Q Plot is even more aligned in the middle. Although the histogram has the central bar slightly higher than the expected, it can still be considered approximately normal.

While the independence assumption is generally supported by the ACF/PACF plots, the Ljung-Box test suggests potential autocorrelation at higher lags. Since independence is a very important assumption, the first model performs better in this respect.

4.4.2. Stationarity and invertibility conditions

Another important part of the validation is checking stationarity and invertibility, since these properties ensure that the model is well defined and behaves properly over time.

If a model is stationary, it can be expressed as an infinite AR process, and if it is invertible, it can be written as an infinite MA process. These representations are useful to understand the dependence structure of the series. Also, because the coefficients become smaller as the lag increases, these expansions can be approximated by truncating them after a sufficiently large lag.

The general expressions are:

$$AR(\infty) : \pi(B)X_t = \frac{\phi(B)}{\theta(B)}X_t = (1 - \pi_1B - \pi_2B^2 - \pi_3B^3 - \dots)X_t = Z_t$$

$$MA(\infty) : X_t = \frac{\theta(B)}{\phi(B)}Z_t = (1 + \psi_1B + \psi_2B^2 + \psi_3B^3 + \dots)Z_t$$

For stationarity, all roots of the AR polynomial $\phi(B)$ must have modulus greater than one. Similarly, for invertibility, all roots of the MA polynomial $\theta(B)$ must also have modulus greater than one.

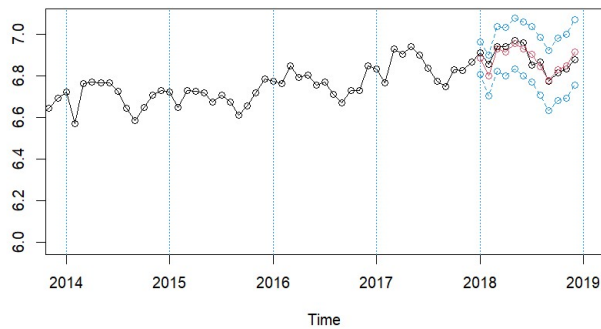
Since these conditions are satisfied for both proposed models, both can be considered stationary and invertible, and therefore admit valid $AR(\infty)$ and $MA(\infty)$ representations.

The computation of the modules of the roots can be found in the annex.

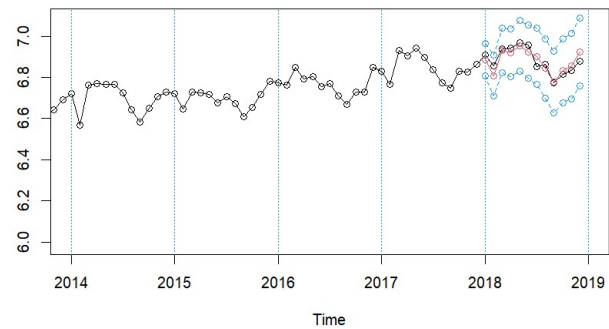
4.4.3. Model stability and accuracy

In order to obtain accuracy measures a model without the last 12 observations was fitted for both models (1 and 2). In both cases the resulting models were stable as the magnitude, sign and significance of the coefficients remained similar to the original models. Stability ensured that the metrics calculated were reliable.

The following plots show the predictions made with the models for the last 12 observations:



(a) Forecasts of the last year for the model 1



(b) Forecasts of the last year for the model 2

The forecast accuracy was evaluated using RMSE, MAE, RMSPE, and MAPE metrics. Both models achieved very low forecasting errors, with MAPE values below 0.4%, indicating high predictive accuracy. Additionally, the width of the confidence intervals was computed as a measure of forecast uncertainty.

The table below shows the metrics of both models:

Model	RMSE	MAE	RMSPE	MAPE	AIC	BIC	AVG_CI
ARIMA(1,1,1)(1,1,1)_12	0.02901799	0.02486329	0.004219325	0.003611575	-1179.529	-1160.459	0.2550604
ARIMA(0,1,2)(0,1,1)_12	0.02982662	0.02610796	0.004336643	0.003793113	-1178.863	-1163.606	0.2607517

Table 1. Accuracy measures. Explicit computation can be found in the annex

Stability was checked in order to be able to extrapolate this metrics to the models with all the observations (the real models).

4.4.4. Model selection

Both models show very similar results. The error metrics for the first model are slightly lower and it has a lower AIC. However, the first model has a better BIC. Additionally, the average width of the confidence intervals for the first model is narrower, suggesting higher precision in its predictions.

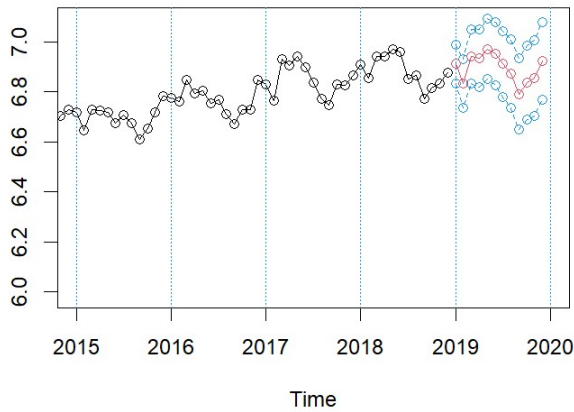
Even though the metrics obtained are very similar, the first model is slightly better in most of the cases. Furthermore, the validation of the residuals (especially the independence) was also clearer and more strongly supported in this model.

4.5. Forecasting

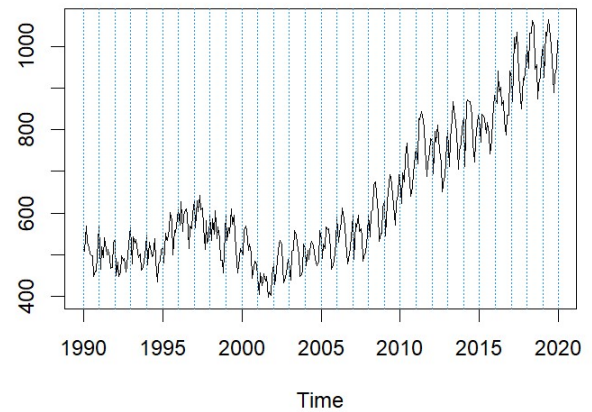
Using the final selected ARIMA model, forecasts for the year 2019 are obtained together with their associated 95% confidence intervals.

Time	Forecast	Lower	Upper
Jan 2019	1003.3492	927.7889	1085.063
Feb 2019	927.4574	841.9208	1021.684
Mar 2019	1034.9344	929.6161	1152.184
Apr 2019	1026.5027	914.9794	1151.619
May 2019	1066.0622	944.1854	1203.671
Jun 2019	1045.3377	920.5575	1187.032
Jul 2019	1003.3745	878.9322	1145.436
Aug 2019	966.9718	842.8030	1109.434
Sep 2019	890.2040	772.1721	1026.278
Oct 2019	932.2277	804.8862	1079.716
Nov 2019	950.0835	816.6332	1105.342
Dec 2019	1015.3506	868.9437	1186.425

Table 2. Forecasts and 95% confidence intervals for 2019



(a) Forecasts and 95% confidence intervals for 2019



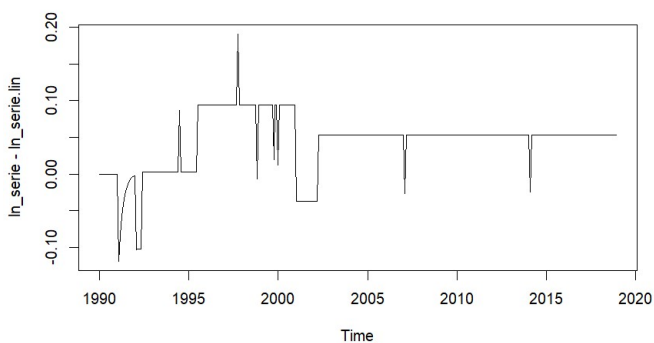
(b) Forecasts back-transformed from the logarithmic scale to the original scale

We can see that the model closely follows the seasonal patterns observed in the historical data. As expected, the model predicts higher renewable energy production during the spring months (March-May) and at the end of the year (December).

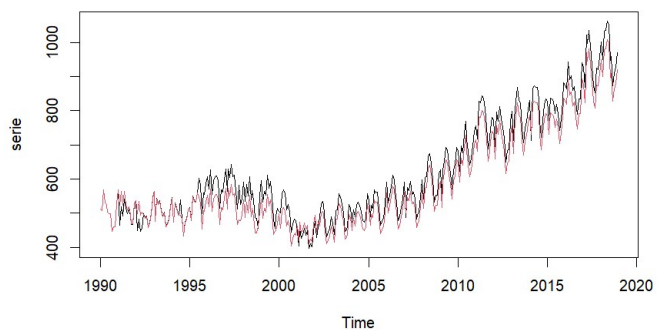
These results suggest that the model can be useful for short-term forecasts of total renewable energy production in the United States. The predictions can help decision-makers plan resources, manage energy supply, and anticipate seasonal changes in demand.

4.6. Outlier treatment

As residual diagnostics suggested the presence of atypical observations, we investigate whether removing outliers can improve the model performance and lead to a better validated model. An automatic outlier detection procedure was therefore applied to the selected model $ARIMA(1,1,1)(1,1,1)_{12}$.



(a) Detected outliers



(b) Original series overlapped with the series after outlier removal

The most significant outlier corresponds to observation 133 (January 2001), classified as a level shift, with an absolute ratio of 3.85. Other relevant interventions were detected around 1997-2002 and during 2007 and 2014, suggesting possible fluctuations in the series.

These observations were corrected through linearization, producing an adjusted series on which the identification stage was repeated. First, the linearized series was transformed into stationary. After this, the ACF and PACF of the series were analyzed.

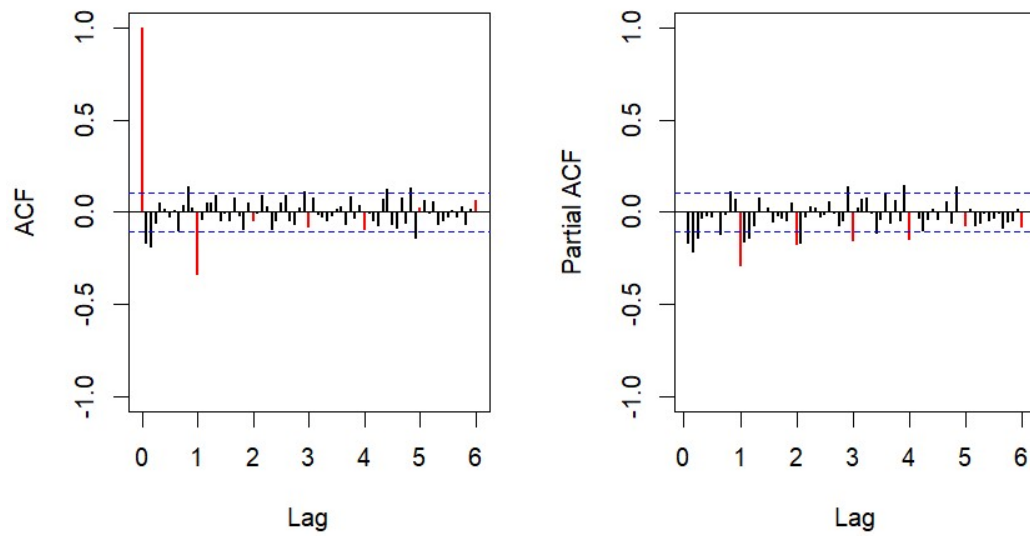


Figure 17. ACF and PACF of the series after outlier removal

After reanalyzing the plots, two candidate models were proposed:

$$ARIMA(1, 1, 1)(0, 1, 1)_{12}$$

$$ARIMA(0, 1, 2)(0, 1, 1)_{12}$$

Based on the AIC criterion, the model $ARIMA(1,1,1)(0,1,1)_{12}$ was selected for validation. After fitting the adjusted model, the residual diagnostics were repeated following the same validation procedure as before.

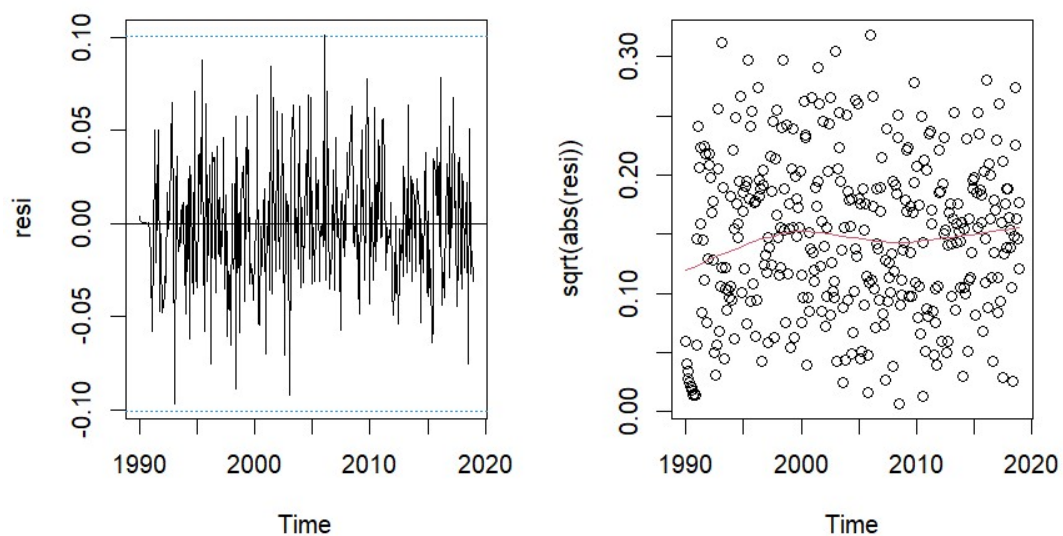


Figure 18. Plot of the residuals of model without outliers

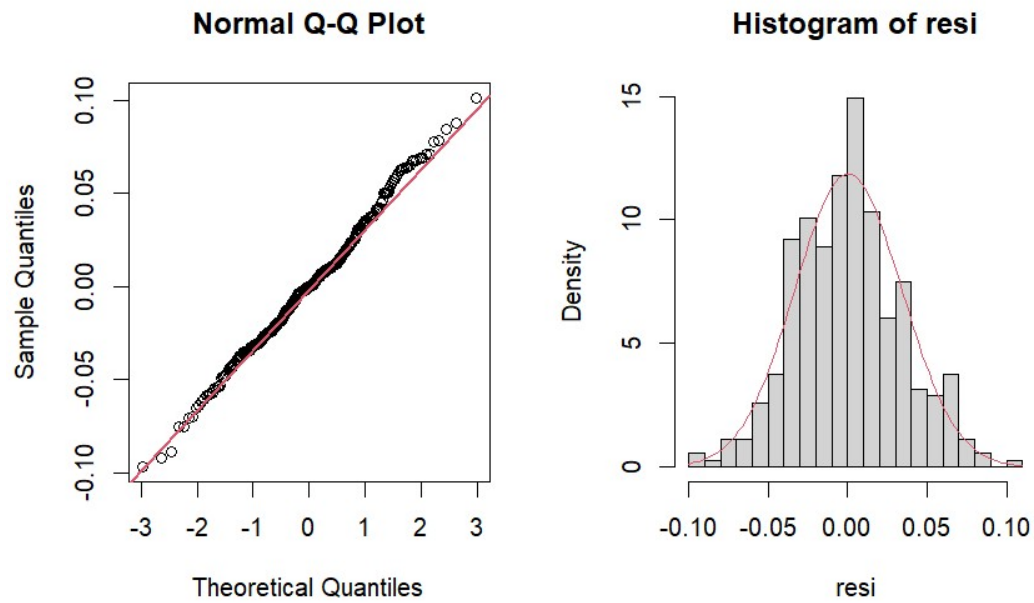


Figure 19. Q-QPlot and Histogram of the residuals of model without outliers

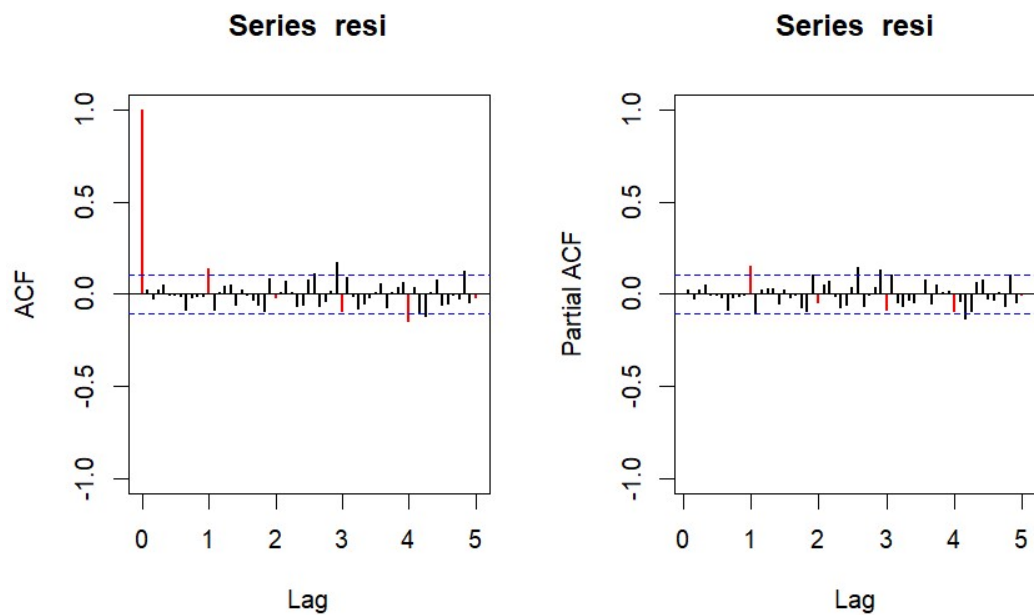


Figure 20. ACF and PACF of the residuals of model without outliers

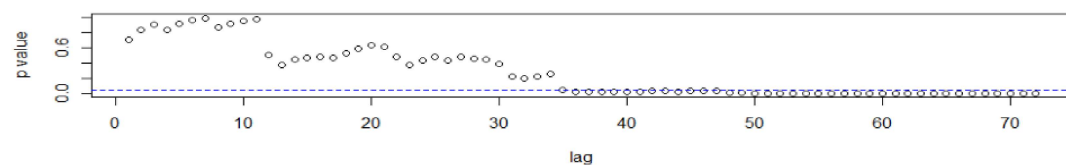


Figure 21. Ljung-Box test plot of model without outliers

Compared with the original model, the corrected specification presents a lower residual variance. Residual diagnostics showed acceptable stationarity and approximate normality conditions. However, the independence assumption could not be properly validated, as the Ljung-Box diagnostics suggest remaining autocorrelation in the residuals.

Therefore, although outlier treatment reduced the variance of the model, it did not improve the overall validity of the model assumptions. Since residual independence is the most important thing to validate, the model without outliers was discarded and the original $ARIMA(1,1,1)(1,1,1)_{12}$ model was retained as the final selected model.

5. CONCLUSIONS

In conclusion, the developed model provides a reliable framework for short-term energy forecasting. By stabilizing the variance with a logarithmic transformation and correcting the regular and seasonal non-stationarity, we have achieved a MAPE of 0.36%, which is a very good metric. This model stands as a valid tool for the U.S. Energy Information Administration to anticipate supply fluctuations and manage the transition toward more sustainable energy resources.

6. REFERENCES

- Course lecture notes: Data Analysis, DSE-UPC 2025-26.
- U.S. Energy Information Administration. (2024). Renewable energy explained: How renewable energy contributes to the U.S. electricity generation (Report). EIA. <https://www.eia.gov/energyexplained/renewable-sources/>
- U.S. Energy Information Administration. (2025). Monthly total primary renewable energy production and consumption, United States, 1990-2019 (Data set). EIA. <https://www.eia.gov/totalenergy/data/browser/index.php?tbl=T10.01#/f=M&start=199001&end=201901&charted=67-8-9-14>

7. ANNEXES

7.1. Annex A: Model selection details

This section presents the parameter estimation results for the candidate ARIMA models considered during the model selection stage. The estimation was carried out using the `arima()` function in R.

7.1.1. $ARIMA(1,1,1)(1,1,1)_{12}$

Parameter	Estimate	Std. Error	t-ratio	Significant
AR(1)	0.4867	0.1248	3.90	TRUE
MA(1)	-0.7605	0.0943	-8.06	TRUE
SAR(1)	0.1752	0.0684	2.56	TRUE
SMA(1)	-0.9103	0.0473	-19.24	TRUE

Table 3. Parameter estimated for Model 1

$$\sigma^2 = 0.001594, \quad \log\text{-likelihood} = 594.76, \quad \text{AIC} = -1179.53$$

7.1.2. $ARIMA(0,1,2)(0,1,1)_{12}$

Parameter	Estimate	Std. Error	t-ratio	Significant
MA(1)	-0.2605	0.0549	-4.74	TRUE
MA(2)	-0.2155	0.0566	-3.81	TRUE
SMA(1)	-0.8307	0.0450	-18.46	TRUE

Table 4. Parameter estimated for Model 2

$$\sigma^2 = 0.001623, \quad \log\text{-likelihood} = 593.43, \quad \text{AIC} = -1178.86$$

7.1.3. $ARIMA(1,1,1)(0,1,1)_{12}$

Parameter	Estimate	Std. Error	t-ratio	Significant
AR(1)	0.4321	0.1221	3.54	TRUE
MA(1)	-0.7260	0.0932	-7.79	TRUE
SMA(1)	-0.8337	0.0453	-18.42	TRUE

Table 5. Parameter estimated for Model 3

$$\sigma^2 = 0.001640, \quad \log\text{-likelihood} = 591.58, \quad \text{AIC} = -1175.16$$

7.1.4. $ARIMA(2,1,0)(0,1,1)_{12}$

$$\sigma^2 = 0.001641, \quad \log\text{-likelihood} = 592.10, \quad \text{AIC} = -1176.21$$

Parameter	Estimate	Std. Error	t-ratio	Significant
AR(1)	-0.2422	0.0544	-4.46	TRUE
AR(2)	-0.2586	0.0536	-4.82	TRUE
SMA(1)	-0.8134	0.0449	-18.10	TRUE

Table 6. Parameter estimated for Model 4

7.2. Annex B: Roots analysis

This section presents the analysis of the roots of the AR and MA polynomials for the selected models. These results are used to verify stationarity (causality) and invertibility conditions.

7.2.1. $ARIMA(1,1,1)(1,1,1)_{12}$

AR polynomial roots (modulus):

$|z| = (1.156226, 1.156226, 1.156226, 1.156226, 1.156226, 1.156226, 1.156226, 1.156226, 1.156226, 1.156226, 1.156226, 2.054746)$

MA polynomial roots (modulus):

$|z| = (1.007865, 1.007865, 1.007865, 1.007865, 1.007865, 1.007865, 1.007865, 1.007865, 1.007865, 1.007865, 1.007865, 1.315006)$

7.2.2. $ARIMA(0,1,2)(0,1,1)_{12}$

AR polynomial: $\phi(B) = 1$ (no AR part)

MA polynomial roots (modulus):

$|z| = (1.015573, 1.015573, 1.015573, 1.015573, 1.015573, 1.015573, 1.015573, 1.015573, 1.015573, 1.015573, 1.015573, 1.633002, 2.841890)$